Volatility Voyage

Assignment 5

Team: **RiskTakers**

Summary & answer to the following Question

1. **Which distribution best models the returns?**

   The Kolmogorov–Smirnov (KS) test indicates that the **Student's t-distribution** provides the best fit for the returns, with a p-value of 0.9591. This is the only distribution for which we fail to reject the null hypothesis that the returns follow the specified distribution. The Cauchy distribution is the next best fit (p=0.5439), but the t-distribution is clearly superior.

2. **What are the estimated parameters?**

   The estimated parameters for the t-distribution are:

   - Degrees of Freedom (ν): 2.7897
   - Location (μ): -0.000499
   - Scale (σ): 0.008489

3. **What are the test results and p-values?**

   Kolmogorov–Smirnov Test Results:

Rank	Distribution	KS Statistic	p-value
1	Student's t	0.0315	0.9591
2	Cauchy	0.0501	0.5439
3	Weibull	0.0915	0.0289
4	Log-normal	0.0922	0.0271
5	Normal	0.0974	0.0165

   Additional normality tests:

   - **Anderson–Darling (Normal):** Test statistic = 3.9392, all critical values exceeded → *Reject normality.*
   - **Jarque–Bera:** JB statistic = 164.08, p-value < 0.0001 → *Reject normality.*
   - **Skewness:** 0.7274 **Kurtosis:** 6.7011

4. **What do the confidence interval outliers indicate?**

   Outliers lying beyond the **95%** confidence band of the fitted distribution may signal unusual return behavior, possibly due to high volatility events or market anomalies. These points are valuable for stress testing and risk management purposes.

Conclusion: The returns are best modeled by a Student's t-distribution with estimated parameters $df \approx 2.79$, $\mu \approx -0.0005$, $\sigma \approx 0.0085$. Goodness-of-fit tests strongly reject normality and favor the t-distribution, which captures the heavy tails and excess kurtosis observed in the data.